

Gold Petal — every strategy, archived

23 August 2026. Formulas as implemented in source. S16 HHHL+wick 1h is the only book that may trade live. Every other book is off live permanently. This PDF does not arm Angel and does not change the S16 formula.

Live size is still yours: tick Lots or ■ on S16, type LIVE, Arm, then RESTART. Leftover Exit still squares an old Angel leftover without Arm. If Angel is already flat, do not Exit.

How to read a candle

O open, H high, L low, C close. Green body = $C > O$. Red body = $C < O$. Doji = $C = O$. Upper wick = $H - \max(O, C)$. Lower wick = $\min(O, C) - L$. HH = this high $>$ previous high. LL = this low $<$ previous low. FLIP = close the open side, then open the opposite. Fill = this bar's close (or LTP in a last-15m daily window). Rank and live P&L are after Angel charges, tax excluded.

Gold Petal session

Mon–Fri 09:00–23:30 IST. S16 is the session/intraday book: flatten at 23:30, leftover at next 09:00. Monthly Gold Petal rollover uses ROLLOVER_DAYS=5.

S16 HHHL + wick 1h — THE ONLY LIVE BOOK

Id: S16_HHHL_WICK_1H

Hold / status: Intraday. Flatten at MARKET_CLOSE (23:30 IST). Leftover at next MARKET_OPEN. Never overnight.

Wait for the 1-hour candle to finish (first tick of the next hour). Compare that closed bar to the previous same-session closed hour. Never use yesterday's last hour or a preopen hour as prev. A restart after the hour already finished still applies the last closed session 1h vs the hour before it (fill at that bar's close), unless a signal was already recorded at or after that close. Fill at this bar's close. FLIP if already the other side. No range skip. No bald-body. No open=high/low (that is S14). min_wick_gap = 0 on the live book. Mood and market regime are observe-only.

Calculation

```
Let c = this finished 1h, p = previous same-session finished 1h.
green = c.C > c.O      red = c.C < c.O
HH = c.H > p.H         LL = c.L < p.L
upper wick U = c.H - max(c.O, c.C)
lower wick L = min(c.O, c.C) - c.L
```

```
If c.C > p.C → HH/LL only (wicks ignored):
    LONG if HH and green
    SHORT if LL and red
    else skip
```

```
If c.C < p.C → wick only (HH/LL ignored), |U-L| ≥ 0:
    LONG if L > U
    SHORT if U > L
    else skip (equal wick)
```

```
If c.C = p.C → skip
```

First finished session hour is stored as prev (no trade).

Next finished session hour can BUY/SHORT vs that prev.

Opposite signal → FLIP (CLOSE then BUY/SHORT).

MARKET_CLOSE → flatten. Next open → leftover flatten if still open.

S1 net-delta (30m pressure)

Id: S1_NETDELTA

Hold / status: Removed from live. Legacy 30-minute net-pressure.

30-minute net-pressure on TBQ/TSQ. Signals BUY / SHORT / CLOSE / reverse. Exhaustion: large price move from entry plus netΔ stretched vs entry netΔ. Divergence: tiny price change but netΔ surges vs previous netΔ.

Calculation

```
NET = TBQ - TSQ over the 30m bar.
```

Default exhaustion: $\text{price} \geq 2\%$ from entry AND $\text{net}\Delta \geq 3 \times \text{entry net}\Delta$.
Default divergence: $|\text{price change}| \leq 0.1\%$ AND $\text{net}\Delta \geq 2 \times \text{previous net}\Delta$.
Not on the live desk.

S2 1-minute depth balance

Id: S2_BALANCE

Hold / status: Removed from live.

Collect buy1..buy5 qty and sell1..sell5 qty for one minute, sum them, then trade on that minute net. Same idea as old tick-flip S2, but once per minute to cut whipsaw.

Calculation

```
net = sum(buy1..5 over the minute) - sum(sell1..5 over the minute)
net > 0 → BUY / hold long
net < 0 → SHORT / hold short
net = 0 → CLOSE / stay flat
```

S3 ML model

Id: S3_ML

Hold / status: Removed from live. Research.

BUY / SHORT / CLOSE from full-depth features (joblib model). Needs a trained model under data/models. Default buy prob 0.58, short 0.42, min hold 30s, every 5 ticks, 80-tick buffer.

Calculation

```
Features from export_full_ticks + ml_features.FEATURE_COLUMNS.
P(up) ≥ 0.58 → BUY
P(up) ≤ 0.42 → SHORT
else HOLD / CLOSE after min hold.
Not on the live desk.
```

S4 daily HH/LL swing (old S13)

Id: S4_OVERNIGHT

Hold / status: Removed from live. Delivery swing. Never next-open.

Old S13 HH+green / LL+red on the day candle, last 15 minutes (23:15–23:30 IST). Holds across days until the opposite HH/LL. FLIP. Lost the Angel/ticks daily backtest to S13 (daily S16). Desk force-off.

Calculation

```
Confirm only in last 15m of this day. Fill at last-15m LTP.
LONG  = day HH and green (H > prevH and C > 0)
SHORT = day LL and red   (L < prevL and C < 0)
Stay until opposite. Monthly rollover flattens last front session.
```

S5 minedge (fee-cover)

Id: S5_MINEDGE

Hold / status: Removed from live. Was delivery overnight.

Trade only when expected move can cover Angel fees. Uses rolling expected-move (ATR + session range) plus strong depth imbalance. Exits on target (+expected), stop ($\sim 0.45 \times \text{expected}$), or flatten.

Calculation

```
expected_points = ATR / session-range estimate
fee break-even ≈ 50 points (■50 round-trip, ■1/point, TURNOVER_MULT=1)
required = max(MIN_EDGE_POINTS, fee_break_even × safety) when COVER_FEES=true
Enter BUY/SHORT only if expected_points ≥ required
AND depth imbalance ≥ ~1.35 (buy vs sell sums)
Target = +expected from entry. Stop ≈ 0.45 × expected.
```

S6 30-point floor

Id: S6_MIN30

Hold / status: Removed from live.

Same family as S5, but the expected-move floor is a flat 30 points.

Calculation

Enter only if $\text{expected_points} \geq 30$. Same imbalance idea as S5.

S8 NET zigzag

Id: S8_NET_ZIGZAG

Hold / status: Removed from live. Was delivery overnight.

Order-book NET zigzags. Long-only while BULL, short-only while BEAR. Entry needs IMB rising-edge or pullback-resume, plus cooldown after exit. After a stop: IMB must fall below min then re-cross.

Calculation

$\text{NET} = \text{TBQ} - \text{TSQ}$

$\text{IMB\%} = |\text{NET}| / \max(\text{TBQ}, \text{TSQ}) \times 100$

Code default $\text{min_imb_pct} = 10$ (some notes said 14 – live code is 10)

$\text{weaken_pct} = 10$ $\text{TP} = 25$ points $\text{SL} = 20$ points

Exit: TP / SL / supporting-qty weaken / bias flip.

S9 30m state

Id: S9_STATE30

Hold / status: Removed from live. Research.

30-minute state machine (journal + optional bar ML). Not on the live desk.

Calculation

State labels from 30m bars. Paper/research only.

S10 legacy 30m zigzag

Id: S10_LEGACY30

Hold / status: Removed from live.

Legacy 30m always-zigzag path (MTF paper experiment).

Calculation

30m zigzag without S8's rising-edge / SL-reset gates.

S11 discovered pack

Id: S11_DISCOVERED

Hold / status: Removed from live. ML-tab hypotheses.

Auto-discovered rules. Paper hypotheses, not proven edge. Off the hot path.

Calculation

Rules written by `discover_strategies.py`. Not live.

S12 30m HH/LL

Id: S12_HHHL30

Hold / status: Removed from live. Retired from the desk.

30m higher-high / lower-low, last-minute confirm. Skips wick-only fakeouts: close must finish beyond the prior high/low.

Calculation

LONG = HH and green and close finishes above prev high

SHORT = LL and red and close finishes below prev low

Last minute of the 30m bar confirms.

S13 daily S16 (day candle)

Id: S13_HHHL_DAY

Hold / status: Removed from live. Was delivery swing. Never next-open.

S16 close-vs-prev formula on the day candle, last 15 minutes (23:15–23:30). Holds the trend across days until the opposite S16 day signal or Gold Petal monthly rollover (flatten last front session, then trade next month). FLIP.

Calculation

Same s16_bar_decision(prev_day, this_day, min_wick_gap=0).
Confirm only last 15m. Fill at LTP in that window.
Rollover_DAYS=5: flatten front month on last front session.

S14 30m wick

Id: S14_WICK30_STRICT

Hold / status: Removed from live. Retired.

Same-candle open=high SHORT / open=low LONG, else raw wick. Weak nearly-equal wicks are skipped. Formula order unchanged.

Calculation

If $O = H \rightarrow$ SHORT (bald top)
If $O = L \rightarrow$ LONG (bald bottom)
else $U = H - \max(O, C)$, $L = \min(O, C) - L$
LONG if $L > U$ and $|U-L| \geq \text{gap}$; SHORT if $U > L$.

S15 30m bald body

Id: S15_WICK30_NOWICK

Hold / status: Removed from live. Retired.

Bald 30m body only, HOLD (no reverse). Fewer trades; no real edge filter.

Calculation

Trade only when the 30m bar has no meaningful wick. No reverse on later bars.

S17 close / high / body grid (research)

Id: S17_CLOSE_HIGH_BODY

Hold / status: Never paper. Never live.

Research grid of close-vs-prev, high-vs-prev, and green/red. Eight cells; leftover (equals) skipped. Books: hhhhl, wick, and, or.

Calculation

1 $C > pC$ $H > pH$ $C > 0$ up_hh_green
2 $C > pC$ $H > pH$ $C < 0$ up_hh_red
3 $C < pC$ $H < pH$ $C > 0$ dn_lh_green
4 $C < pC$ $H < pH$ $C < 0$ dn_lh_red
5 $C < pC$ $H > pH$ $C > 0$ dn_hh_green
6 $C > pC$ $H < pH$ $C < 0$ up_lh_red
7 $C > pC$ $H < pH$ $C > 0$ up_lh_green
8 $C < pC$ $H > pH$ $C < 0$ dn_hh_red
hhhl = S12 sides. wick = raw wick. and = both same. or = either, skip fight.

S18 OHLC + volume + yesterday 1h

Id: S18_OHLC_VOL_HTF

Hold / status: Removed from live. Was delivery overnight.

Wait for the 1h to finish. Need previous same-session 1h and yesterday's completed day. Bar volume is the delta of Angel session-cumulative volume_trade_for_the_day. Learner may replace the pack from ticks. Base book is AND of every leg.

Calculation

Base LONG when ALL are true:
 $C > 0$, $C > \text{prev}C$, $H > \text{prev}H$, $\text{vol} > \text{prev vol}$, $C > \text{yesterday } C$
Base SHORT when ALL are true:
 $C < 0$, $C < \text{prev}C$, $L < \text{prev}L$, $\text{vol} > \text{prev vol}$, $C < \text{yesterday } C$
Anything equal / missing prev or day / volume not up $\rightarrow$ skip.
FLIP at this bar's close. Fill at close.

S19 aligned body + close 1h

Id: S19_BODY_CLOSE_1H

Hold / status: Removed from live. Was delivery overnight.

Wait for the 1h to finish. Need previous same-session 1h. LONG when green AND close up vs prev. SHORT when red AND close down. Mixed / doji / equal close: hold. FLIP only on the opposite aligned hour. 100-lot + fees 1h backtest lost after charges vs S16 and S18.

Calculation

LONG = $C > 0$ and $C > \text{prevC}$

SHORT = $C < 0$ and $C < \text{prevC}$

Skip mixed (green but $C < \text{prevC}$, red but $C > \text{prevC}$), doji ($C=0$), $C=\text{prevC}$, missing prev.

Hold through skip. FLIP only on opposite aligned signal. Fill at close.

S20 fade low / high 1h

Id: S20_FADE_HL

Hold / status: Removed from live. Never armed.

Buy a bounced low, short a rejected high. Inside / outside / knife (LL that closes red) / chase (HH that closes green): hold. 1h after-charges backtest lost to S16 and S18.

Calculation

LONG = $L < \text{prevL}$ AND not $H > \text{prevH}$ AND $C > 0$

SHORT = $H > \text{prevH}$ AND not $L < \text{prevL}$ AND $C < 0$

FLIP only on the opposite fade. Fill at this bar's close.

Overnight gap (close → next open)

Id: OVERNIGHT_GAP

Hold / status: Removed from live. Was next-open exit, never EOD flatten. Mood-exempt.

Read today's Gold Petal tape near MARKET_CLOSE. BUY if the read says tomorrow's open is up; SHORT if down; skip if mixed. CLOSE in the first 5 minutes after next MARKET_OPEN (09:00–09:05 IST). Not S4 daily swing.

Calculation

Need ≥ 20 ticks, valid OHLC.

$\text{day_ret} = \ln(C / O)$

$\text{CLV} = 2 \times (C - L) / (H - L) - 1$ (+1 close at high, -1 at low)

$\text{late_ret} = \ln(C / \text{late_open})$ over last 45 minutes

$\text{late_imb} = (\text{TBQ} - \text{TSQ}) / (\text{TBQ} + \text{TSQ})$ late window

$z = 6 \cdot \tanh(\text{day_ret} / 0.004) + 1.2 \cdot \text{CLV} + 4 \cdot \tanh(\text{late_ret} / 0.002) + 0.9 \cdot \tanh(\text{imb} / 0.35)$

$P(\text{up}) = \text{sigmoid}(z)$

BUY if $P \geq 0.58$ SHORT if $P \leq 0.42$ else skip

Entry window: last 15 minutes before 23:30. Exit: 09:00–09:05 next session.

FLOW_BRAIN tick pressure

Id: FLOW_BRAIN

Hold / status: Removed from live. Never armed.

Every second: classify LTP + TBQ/TSQ change (new buy qty vs new sell qty). Trade only continuation + expansion. Flatten on decay / opposite / EOD. Desk pack also reused S5 fee-cover, S8 NET/IMB, S16/S19 1m structure, S20 fade-block.

Calculation

$\text{bull_cont} = \text{price up} + \text{buy flow up}$

$\text{bear_cont} = \text{price down} + \text{sell flow up}$

Expansion = price response growing AND flow imbalance growing.

Trade continuation+expansion only. Fill at LTP. Rank after charges, tax excluded.

Body-shrink fade (research, no ENABLE)

Id: BODY_SHRINK_FADE

Hold / status: Never a paper/live book. Do not rewrite S16.

Three candles: two same-color bodies, second smaller and lower/higher, fade candle 3.

Calculation

Research overlay only. python backtest_body_shrink_fade.py

AMISE slots S21–S24

Id: AMISE_S21_S24

Hold / status: Removed from live. Research factory.

Lab challengers after Approve. Off the live desk list.

Calculation

Factory books. ENABLE_S21..S24 default false. Not live.

What was removed from live

S1, S2, S3, S4, S5, S6, S8, S9, S10, S11, S12, S13, S14, S15, S17, S18, S19, S20, overnight gap, FLOW_BRAIN, AMISE S21–S24, body-shrink fade. Source files stay in the repo as this archive's reference. They cannot be armed. The Live table lists S16 only.

S16 checklist (unchanged formula)

- Wait for the 1h to finish.
- Prev must be the previous same-session hour.
- Up-close → HH+green LONG / LL+red SHORT.
- Down-close → lower wick LONG / upper wick SHORT (gap 0).
- Equal close → skip.
- FLIP at that close. Flatten at MARKET_CLOSE.
- You still Arm live. The desk does not.

End of archive. Keep this PDF. The live desk now trades S16 only.